



Magic Leap raises \$1.4B

Magic Leap, the augmented reality startup acquires a 3D division of Dacuda, a computer vision startup of Zurich. <http://dgit.in/McLeap>



No cheap HTC phones

The Taiwanese smartphone maker announces that it will exit the entry-level smartphone market later this year. <http://dgit.in/HTCcheap>



Beaming Kazakh Students with their AI-Farabi-1 Satellite - on top of the world, already!

days before it struck the Philippines in 2013. Described as one of the "most intense tropical cyclones on record", evacuation and rescue efforts can be geared up accordingly. ISRO also holds the distinction of operating the world's largest constellation of remote sensing satellites – 15 satellites, serving the nation from the SSO.

The PSLV has a 4-stage design. Broadly speaking, it includes four groups of rocket engines operating in succession to propel the payload into space. When the Stage I engine completes operation, it detaches & falls off with Stage II kicking in, and so on. This ensures no dead weight travels upwards and each stage of the flight can be powered by an engine suited to that flight profile. While the second and fourth stage of the PSLV is liquid-powered, the traditional solid fuel continues powering the first and third stage.

Satellites

The ISRO-built Cartosat-2D satellite was the mission's queen bee. Weighing 714 kg, its cameras are capable of capturing clear images at a resolution of 0.65 m. Announced prior to its launch, through its five year design-life, its imagery serve for cartographic, urban and rural, coastal land use and regulation, utility management like road monitoring, water distribution, creation of land

use maps, geographical and man-made features detection, and various others including Land Information Systems (LIS) and Geographical Information System (GIS) applications.

Other than the Cartosat-2D satellites, there are two additional Indian satellites – the INS-1A & INS-1B observation satellites. Designed to last six months, they have been built upon a completely new bus system – the ISRO Nano Satellite (INS) bus. A satellite bus system is like an application SDK, providing the physical & specification framework around which builders can develop their application-specific satellites. Besides simplifying satellite-building, they introduce standardisation to further simplify the launch process. Traditional ISRO developed bus systems have been geared towards satellites weighing above 1 metric tons – I-1K, I-2K, I-3K & I-4K. New fabrication technologies have miniaturised components to an extent where satellites weighing tonnes have now scaled down to kilograms, exhibiting better performance. In fact, the Google Nexus One smartphone was powerful and fast enough to be used on the British STRaND-1 satellite launched by, you guessed it, ISRO in 2013.

Availability of a system like the INS Bus would go a long way in encouraging greater participation by R&D labs

in space activities. They now have a platform to devise their own interesting experiments. No longer will they be dependent on big satellite builders to allocate some of the sparse on-board real-estate. To put things into perspective, one of the microgravity experiments designed on the Swiss DIDO-2 Nano-satellite, also known as CHEN Jiayong-1, is studying the "formation of a drug in a microfluidic device". This will ultimately lead to the development of 3D-printed medicines in future. In the foreseeable future, it could become a common occurrence for privately managed PSLV launches to deploy scores of such Nano-satellites every month. Aims to democratise space just got a big boost.

All said and done, this record is solely attributed to the launch requirements of one participating American company – Planet Labs Inc. 88 of their Flock-3P Nano-satellites or 'Doves' were part of the mission, now orbiting the Earth. With its constellation of 100 operational Doves, it now claims the ability to photograph the entire Earth every day. This San Francisco-based company in fact, is an old customer. During last year's PSLV-C34 mission, ISRO had launched Planet Labs' first 12 satellites.

The other American Company, Spire Global, Inc., too has returned to ISRO to launch its eight LEMUR satellites.

PSLV Variants and Payload capability					
PSLV VARIANTS	PSLV	PSLV-CA	PSLV-XL	PSLV-HP	PSLV-3S
	BASE CONFIGURATION	CORE ALONE VEHICLE, NO STRAP-ONS	PSLV WITH EXTENDED STRAP-ON (PSOM-XL)	PSLV-XL WITH HIGH PERFORMANCE PS4	PS2 & STRAP-ONS REMOVED
	(S59 + S139) + PL40 + HP33 + PS4	S139 + PL40 + HP33 + PS4	(BS12 + S139) + PL40 + HP33 + PS4	(BS12 + S139) + PL40 + HP33 + PS4	S139 + HP33 + PS4
	LEO	3200	2100	3800	4000
SSPO (822 Km)	1600	1100	1750	1900	-
GTO (246 x 24000 km)	1150	-	1300	1440	-